

OS Unit 5 Question Bank — File Systems & I/O

Operating Systems (OS) · Unit 5 · File Systems & I/O · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. Define File Concepts. Derive or justify the key properties involved.
2. Explain Directory Structures with a neat diagram and a suitable example.
3. Define Allocation Methods. Derive or justify the key properties involved.
4. Write short notes on Free Space Management.
5. What is meant by Disk Scheduling? Discuss its advantages and limitations.
6. Illustrate RAID with a worked example.

Part B — Long answer (10 marks)

1. Discuss the role of File Concepts in File Systems & I/O.
2. Explain Directory Structures with a neat diagram and a suitable example.
3. Discuss the role of Allocation Methods in File Systems & I/O.
4. What is meant by Free Space Management? Discuss its advantages and limitations.
5. Illustrate Disk Scheduling with a worked example.
6. Discuss the role of RAID in File Systems & I/O.
7. What is meant by File Concepts? Discuss its advantages and limitations.
8. What is meant by Directory Structures? Discuss its advantages and limitations.
9. Write short notes on Allocation Methods.
10. Compare Free Space Management with Disk Scheduling. When would you prefer one over the other?
11. Explain Disk Scheduling with a neat diagram and a suitable example.
12. Write short notes on RAID.